

PhD Student in Statistics, Stanford University   Stanford, CA  
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|-----------------------------------------------------------------|------------------------------------|
| <b>Stanford University</b> , Stanford, CA, United States        | 09/2025 – Present                  |
| Ph.D. in Statistics                                             | Cumulative GPA: 4.05               |
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| <b>McGill University</b> , Montreal, QC, Canada                 | 09/2021 – 12/2024                  |
| Bachelor of Science in Honours Mathematics and Computer Science | Cumulative GPA: 4.00               |
| Minor in Statistics                                             | Graduated with First-Class Honours |

My research interests lie in statistics and machine learning. I am particularly interested in uncertainty quantification, reliable AI/ML, and statistical inference with black-box models. My current work focuses on conformal inference, with broader interests in causal inference, empirical Bayes, and multiple testing.

6. [Bai T](#), [Jin Y](#). **Conformal Selective Prediction with General Risk Control**. *Under review at the Journal of the American Statistical Association*, 2026.  
*Summary*: Introduces a general conformal framework for selective prediction that provides finite-sample, distribution-free control for arbitrary risk measures via multiple testing with generalized e-values.
5. [Shi W](#), [Li Y](#), [Bai T](#), [Zhang Y](#), [Zhao K](#), [Legault MA](#), [Peng H](#), [Zhao Y](#), [Kolaczyk ED](#), [Yu X](#), [Yang A](#). Beyond Go/No-Go Decisions: A Regional Selection Framework for Uncertainty-Aware Molecule Screening. *ChemRxiv preprint*, 2026.  
*Summary*: Develops a flexible three-way, uncertainty-aware framework for compound screening that moves beyond binary go/no-go decisions.
4. [Bai T](#), [Tang P](#), [Xu Y](#), [Svetnik V](#), [Khalili A](#), [Yu X](#), [Yang A](#). Conformal Selection for Efficient and Accurate Compound Screening in Drug Discovery. *Journal of Chemical Information and Modeling*, 2025.  
*Summary*: Applies conformal selection to compound screening, delivering distribution-free error control together with improved screening efficiency and accuracy.
3. [Bai T](#), [Zhao Y](#), [Yu X](#), [Yang A](#). **Multivariate Conformal Selection**. *Forty-Second International Conference on Machine Learning (ICML 2025)*.  
*Summary*: Extends conformal selection to multiple response variables, providing finite-sample risk control under complex, multidimensional selection criteria.
2. [Bai T](#), [Jin Y](#). **Optimized Conformal Selection: Powerful Selective Inference After Conformity Score Optimization**. *Under review (minor revision) at the Journal of the Royal Statistical Society: Series B*, 2024.  
*Summary*: Enables data-driven conformity-score optimization and model selection prior to candidate selection while preserving distribution-free FDR control and improving statistical power.
1. [Grant L](#), [Diagne M](#), [Aroutiunian R](#), [Hopkins D](#), [Bai T](#), [Kondrup F](#), [Clark G](#). Machine learning outperforms the Canadian Triage and Acuity Scale (CTAS) in predicting need for early critical care. *Canadian Journal of Emergency Medicine*, 2024.  
*Summary*: Uses interpretable machine learning to predict early critical-care needs in emergency departments, outperforming the Canadian Triage and Acuity Scale.

- Conformal Selective Prediction with General Risk Control
  - ICSA Applied Statistics Symposium, Student Paper Award Session 06/2026
  - Berkeley-Stanford Joint Colloquium, 2026 04/2026
- Model-Free Selection Inference for Drug Discovery via Conformal Prediction
  - McGill Seventh Undergraduate Research Conference 08/2024

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|-----------------------------------------|---------------------|------|
| William R. and Sara Hart Kimball Fellow | Stanford University | 2025 |
| ICSA Student Paper Award                | ICSA                | 2026 |
| ICSA Travel Award                       | ICSA                | 2026 |
| First-Class Honours                     | McGill University   | 2025 |
| Charles Fox Memorial Prize              | McGill University   | 2025 |

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|-----------------------------------------------------------|-------------------|------------------|
| Science Undergraduate Research Award (SURA)               | McGill University | 2024             |
| Sir Edward Beatty Memorial Scholarship in Mathematics     | McGill University | 2024             |
| Tomlinson Engagement Award for Mentoring                  | McGill University | 2023, 2024       |
| A. D. Pelletier Scholarship in Mathematics and Statistics | McGill University | 2023             |
| Robert W. Wilson Scholarship                              | McGill University | 2022             |
| Dean's Honour List                                        | McGill University | 2022, 2023, 2025 |

## TEACHING

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### Stanford University

- Teaching Assistant, [DATASCI 112: Principles of Data Science](#) Winter 2026

### McGill University

- Teaching Assistant, [MATH 350: Honours Discrete Mathematics](#) Fall 2023
- Teaching Assistant, [MATH 457: Honours Algebra 4](#) Winter 2024
- Mathematics Helpdesk Tutor Fall 2023
- Computer Science Helpdesk Tutor Winter 2024

## SERVICE

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- Journal Reviewing: TPAMI, Statistics and Computing
- Conference Reviewing: NeurIPS 2026

## WORK EXPERIENCE

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### Vffice, Inc.

*Software Developer Intern*

Longueuil, QC, Canada  
05/2023 – 08/2023

Developed and integrated custom business management solutions using Microsoft Azure and C#.

## EXTRACURRICULAR ACTIVITIES

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- Competitive Programming: Contestant with [Compete McGill](#). Team ranked 5th in the 2023 NAQ.

## SKILLS

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- Programming: Python (PyTorch, scikit-learn), R, C/C++
- Software/Tool:  $\text{\LaTeX}$ , Git, PowerShell, RStudio